

Sensor Timing Firmware

The Teensy 4.1 firmware separates physical data acquisition from timing I/O. The Linux host continues acquiring all four cameras, both VLP-16s, and the Xsens MTi-30 continuously. The firmware can qualify a common reference edge, schedule future trigger pulses, observe per-camera exposure and IMU sync edges, and publish timing diagnostics. It neither transports payload data nor proves that a host driver accepted a hardware trigger.

All V6 timing pins are assigned in `firmware_config.h`, but every firmware timing gate is false. On every disabled boot the firmware probes the battery-backed DS3231 and requests `DS3231_OFF`, so a square-wave mode retained from earlier operation is not silently trusted. That shutdown cannot suppress an edge before MCU initialization and cannot be confirmed if the RTC is unreachable; `FIELD_VALID` hardware output gating and commissioning scope tests remain necessary. A build becomes actuating only after the corresponding polarities, line drivers, configuration, and wiring-verification gates are supplied. The scheduler then requires stable reference periods before running, stops on a missing or out-of-range reference, rejects late trigger deadlines, requires camera ExposureActive feedback, and latches faults rather than continuing open-loop. The TPS3897 `FIELD_VALID` input is another required gate for every field-timed path. While it is LOW, firmware suppresses LiDAR reference/NMEA work, drops camera/MTi timing events, keeps scheduled field outputs inactive, and reports a `field_power_invalid` timing fault and discards the relative epoch. U3/U6 OE gates driven by `FIELD_VALID` provide the primary hardware block without adding a component to any signal data path; firmware handling is secondary. U11 eFuse EN/UVLO may remain on down to roughly 4.15 V because of internal hysteresis; it is therefore not the runtime proof that AHCT field logic remains above its 4.5 V guaranteed range. The separate supervisor qualifies the rail near 4.7 V. Relative timing remains unqualified until the assembled board is measured with an oscilloscope or logic analyzer. V6 pulls the DS3231 open-drain `INT/SQW` output to 3.3 V; its falling edge triggers a hardware one-shot. The resulting active-high pulse fans to both VLP-16s, and Teensy D12 captures that same shaped rising edge. The DS3231 datasheet places the 1 Hz square-wave high transition approximately 500 ms after the seconds-register transfer; capturing the falling edge therefore supports association with the next divider boundary, but it is not proof of exact phase. Scope the assembled module `SQW` against RTC register rollover and the shaped PPS before making any phase claim. A module's 5 V pull-up must never reach the non-5-V-tolerant Teensy. The RTC/one-shot is one shared physical source, not a crosspoint. Enabling it for camera or MTi timing also physically drives both VLP PPS connectors while `FIELD_VALID` is high. `kLidarTimingEnabled` controls LiDAR-specific firmware capture and diagnostics; it is not an independent hardware PPS disconnect. Consequently both VLP PPS branches must be safely wired and electrically commissioned before any common timing function is enabled. The currently available square wave is not a verified UTC/GNSS phase source. Published firmware time references use a monotonic epoch beginning at the first qualified edge and explicitly do not claim UTC. On the existing `sensor_msgs/TimeReference` diagnostics, `header.stamp` is only the ROS publication-receipt time and `time_ref` is the unmapped relative MCU epoch. These topics are non-authoritative and are not a ROS-time calibration contract.

Electrical boundary

The installed-camera inventory must be checked against the physical labels. The current Teledyne reference names the official color model `FG-PGE-50S5C-IP`; the repository's `FG-PGE-50S5C-C-IP` spelling is not assumed to be a second electrical variant.

For the current Forge IP67 family, the M8 eight-pin GPIO interface assigns pin 1 to 8-24 V camera input power, pin 2/Line0 to the opto-isolated input, pin 3 to opto ground, pin 6/Line1 to the opto-isolated output, and pin 7 to camera power ground. Published typical thresholds are 1.55 V low and 2.6 V high for Line0; Line1 is specified as approximately 0.56 V low and 5 V high under the vendor's measurement conditions. The Teensy accepts only 0-3.3 V inputs and its normal output recommendation is 4 mA. Therefore neither Forge line connects directly to the MCU. The board must provide characterized, 3.3 V-safe buffered inputs and outputs and defined returns. `OPTOOUT` is open collector, so each camera gets a separate starting-value 1.0 kOhm pull-up to board 3.3 V referenced to `OPTOGND`; cable rise time and sink current must be measured. V6 carries only M8 pins 2, 3, and 6 for each camera, does not carry camera-power ground pin 7, and does not power the PoE cameras. Touching 1x6 J5 carries cameras 1-2 and 1x6 J7 carries cameras 3-4; each camera's `OPTOGND` is bonded to board signal ground. Use IP-rated M8 cables or sealing plugs to retain the camera's ingress rating.

V6 uses one D11 command and four hardware-buffered fanout branches; it does not claim four sequential GPIO writes are simultaneous. R66-R69 are separate 10 kOhm connector-side pulldowns that hold `CAM1_OPT0IN` through `CAM4_OPT0IN` low during MCU reset/high-Z, core-off, driver-disable, or an open harness. They are static fail-low loads, not RC timing filters. Characterize branch skew on the assembled board. Configure each camera separately for `FrameStart`, the selected Line0 rising edge, and Line1 `ExposureActive`; both device-side functions have separate false-by-default firmware gates. Verify every accepted trigger and exposure edge on the assembled harness before enabling the firmware gate. The current host profile remains free-running and unchanged.

The VLP-16s acquire continuously. Their input is a clock/phase reference, not a scan trigger. The official manual requires a PPS high above 3.0 V and below 5.0 V, low below 1.2 V, and at least 2 mA source current. NMEA and PPS must alternate; PPS-to-NMEA separation is at least 50 ms and the NMEA sentence must finish at least 300 ms before the next PPS. The current firmware has no verified UTC/GNSS source. Its optional nonblocking Serial1 path emits a checksum-correct status-V GPRMC sentence from the local RTC only after the wiring, polarity, RTC state, and measured register-rollover-to-SQW-to-PPS phase are commissioned; it makes no GNSS-position or UTC-traceability claim. The status remains `v`, with blank position fields; firmware must never fabricate receiver-valid status `A`. On each VLP, read back and save both the PPS Qualifier and GPS Qualifier `Require GPS Receiver Valid` settings as OFF before enabling the RTC-only stream. The manual shows OFF as a default, but an assumed default is not commissioning evidence. Verify the delayed PPS Locked state, packet PPS status 2, and copied RMC/time continuity independently on both units. The Teensy UART peripheral, not an extra inverter IC, owns the VLP-required TX polarity inversion; the board's AHCT fanout is non-inverting. Firmware starts a sentence 100 ms after the shaped PPS leading edge. With the nominal 10 ms one-shot, this preserves about 90 ms from PPS trailing edge to NMEA start instead of relying on the 50 ms minimum. It suppresses the sentence if the main loop misses the 200 ms start deadline, aborts a pending tail after the 500 ms enqueue deadline, and feeds only currently available UART-buffer capacity. A conservative 80-byte, 9600-baud bound is 83,334 us, so 500,000 + 83,334 us still leaves the manual's final 300 ms quiet window even at the firmware's minimum accepted 900 ms reference period. The sentence contains the next PPS second, including calendar rollover, because Rev-F describes the LiDAR applying the received time at that next leading edge. Scope verification is mandatory because enabling a second hardware inversion would silently restore the wrong polarity. Dual-VLP phase locking additionally requires a locked PPS, an RPM multiple of 60 between 300 and 1200, and distinct measured phase offsets configured and saved on the sensors. The one-shot and common electrical fanout, not two MCU writes, create the two LiDAR PPS branches.

Live USB and udev inspection on 2026-08-26 confirmed the installed device identifies as an Xsens MTi-30 AHRS. Its serial identity remains host inventory, not an electrical interface requirement. The existing USB connection continues to carry IMU data; V6 adds only the separate SyncIn, SyncOut, and signal-return timing connector. R65 is a 10 kOhm connector-side pulldown that holds `MTI_SYNCIN` low during MCU reset/high-Z, core-off, driver-disable, or an open harness. The encased MTi-30 synchronization contract accepts 0-0.8 V low and 2.5-20 V high on SyncIn/ClockSync; SyncOut is 0-0.4 V low and above 2.9 V high. Firmware cannot determine the selected MTi `SyncSettings`. A 10 kOhm receiver-side pulldown makes an open SyncOut cable deterministic low before the board's inverting Schmitt receiver, so the MCU input idles high rather than floating. The optional MCU output is only a time-event marker for the continuously sampled IMU; it must not select `StartSampling` or reduce navigation data to the camera's current 5 Hz cadence. It therefore remains disabled until polarity, voltage, cable pin, the SyncIn event, the SyncOut feedback event, and a corresponding packet marker are independently verified in MT Manager.

The Imagenex DT100 and Ping360 are intentionally not connected to V6 in this revision. That is a safety boundary, not a statement that hardware sync is impossible. The DT100 has Sync IN/OUT circuitry, but a vendor report identified damage after power was applied to those lines. Do not drive either DT100 sync pin until the exact connector, direction, polarity, voltage levels, and common-reference requirements are vendor-verified. The current safe contract uses host/driver timestamps for both sonars and fuses their asynchronous measurements against a continuous-time trajectory; it does not label them hardware-synchronized.

Firmware structure and validation

`sensor_sync.h` owns the side-effect-free state machine, bounded edge mailboxes, and continuous rollover-safe relative-epoch mapper. GPIO ISRs only deposit timestamped edges; the timer ISR is the sole owner of scheduler, feedback, fault, and relative-time state. Mailbox overflow latches synchronization off. `sensor_sync_runtime.h` owns Teensy GPIO, camera exposure

midpoint capture, IMU events, feedback deadlines, and safe inactive outputs. `firmware_pin_contract.h` rejects cross-owner aliases, RTC-I2C pin collisions, unsupported telescope PWM assignments, and invalid digital pins before either runtime initializes. `main.ino` owns ROS publication, RTC edge labelling, the bounded nonblocking PPS-to-NMEA state machine, the timer ISR, and the disabled telescope runtime.

The pure scheduler is suitable for non-actuating host compilation and state transition checks. These checks are performed from an ephemeral harness rather than a repository validator. They check logic only. A valid Teensy build requires a Teensyduino core, rosserial headers, and RTCLib. Acceptance additionally requires a current-limited bench supply, no motor or sensor payloads connected for initial bring-up, safe inactive outputs during boot/reset/disconnect, instrumented edge-to-edge skew and jitter, deliberate missing-reference and feedback-fault injection, MTi packet-marker verification, VLP PPS-lock/phase-lock status, and read-back of both VLP qualifier pairs with Receiver Valid requirements disabled, packet PPS status 2, copied status-V RMC/time continuity, and camera exposure/frame-counter correlation. Passing simulation or host tests is not physical synchronization evidence.

The telescope motor driver also has external hard pulldowns and a common default-low enable so reset, boot, disconnected MCU, or high-impedance GPIO cannot energize either half-bridge. The independent hard E-stop remains in the 12 V motor-power path. Firmware fail-closed states supplement those electrical controls; they do not replace them.

Primary references

- Teledyne FLIR, [Forge 5 MP GigE camera family](#), model and acquisition-feature reference.
- Teledyne FLIR, [Forge IP67 GPIO electrical characteristics](#), M8 pinout, thresholds, delays, and line ratings.
- Teledyne FLIR, [Forge IP67 acquisition control](#), FrameStart trigger-source/mode/overlap contract.
- Ouster (Velodyne), [VLP-16 User Manual, 63-9243 Rev. F](#), PPS/NMEA electrical, timing, and phase-lock requirements.
- Analog Devices, [DS3231 datasheet](#), open-drain `INT/SQW` electrical contract.
- Xsens, [MTi 10-series and MTi 100-series User Manual, MT0605P](#), encased MTi connector, SyncIn/SyncOut, and ClockSync electrical characteristics.
- PJRC, [Teensy 4.1 technical specifications](#), MCU GPIO voltage and current limits.